

Aiken validator coverage — rejection points and mutation testing

Run: 2026-09-22T02:22:24.103Z-02:22:37.552Z · aiken v1.1.21+42babe5 · 102 tests, 102 green.

Consolidates two working documents that were previously published separately — `mutation-report.md` and `expect-trace-report.md` — into a single file.

Why this is a separate report

Aiken does not measure line coverage the way the TypeScript stack does. The four TypeScript parts (`apps/api`, `apps/web`, `packages/shared`, `packages/cardano`) use `vitest`'s `v8` provider, which instruments every line, branch, statement and function and reports a percentage — that is what `test-report.md` covers. Aiken's toolchain has no equivalent: `aiken check` runs the test suite and, separately, a `--property-coverage` flag exists but does not produce a line percentage.

So the validator's coverage evidence is built with two purpose-written scripts (`contracts/scripts/rechazos-mutantes.mjs` and `contracts/scripts/rechazos-trazas.mjs`, documented in `contracts/CLAUDE.md`) that answer the same question — "is every rejection path actually exercised by a test?" — with methods that fit how a Plutus validator fails:

- 1. **Rejection point** → **test**. Every `expect` in the validator is a point where a malformed or invalid transaction aborts. For each one, the table below names the test(s) that trigger it.
- 2. **Mutation testing**. For every other check that rejects a transaction (an `and` condition, a boolean `expect`, a list pattern, a `when` branch, a `fail`), the script removes or weakens that one check, reruns the suite, and requires at least one test to turn red. A check that can be deleted without failing any test is a check nothing actually defends.

Both scripts run against `contracts/validators/stage.ak` and `contracts/lib/propnexus/fsm.ak` (the stage-transition validator and the shared finite-state machine it enforces), and are **not** run in CI — each mutant re-runs `aiken check`, which takes several minutes for all 51. They are run by hand and their output is what this document records.

Rejection point → test

18 `expect` statements: 18 with a test, 0 without.

Every `expect` in the validator is a point where the transaction aborts if the pattern it expects does not match. This table maps each one to the test(s) that make it abort exactly there.

File:line	expect	Tests that abort there
<code>validator/s/stage.ak:69</code>	<code>expect Finite(lower) = range.lower_bound.bounds_type</code>	<code>spend_rejects_open_ended_validity_range</code> , <code>spend_rejects_missing_lower_bound</code>

File:line	expect	Tests that abort there
validator s/stage.akk:70	expect Finite(upper) = range.upper_bound.bound_type	spend_rejects_missing_upper_bound
validator s/stage.akk:84	expect Some(old_datum) = datum	spend_rejects_missing_datum
validator s/stage.akk:85	expect Some(own_input) = find_input(tx.inputs, own_ref)	spend_rejects_unknown_own_ref
validator s/stage.akk:87	expect Script(own_policy) = own_address.payment_credential	spend_rejects_own_input_not_locked_by_a_script
validator s/stage.akk:91	expect assets.quantity_of(own_input.output.value, own_policy, old_datum.stage_ref,) == 1	spend_rejects_utxo_without_thread_token, spend_rejects_thread_token_of_another_stage, spend_rejects_utxo_with_two_units_of_own_token, spend_rejects_own_input_with_two_units_when_a_decoy_absorbs_carrying_thread
validator s/stage.akk:100	expect inputs_at_address(tx.inputs, own_address) == 1	spend_rejects_two_script_inputs
validator s/stage.akk:101	expect [continuing_output] = at_address(tx.outputs, own_address)	spend_rejects_a_second_output_at_the_exact_script_address, spend_rejects_two_script_outputs, spend_rejects_no_continuing_output
validator s/stage.akk:104	expect [_] = carrying_thread(tx.outputs, own_policy, old_datum.stage_ref)	spend_rejects_thread_split_across_a_staking_variant_of_the_script_address, spend_rejects_value_drain, spend_rejects_dropping_the_thread_token
validator s/stage.akk:107	expect list.has(tx.extra_signatories, admin)	spend_rejects_missing_admin_signature, mint_rejects_missing_admin_signature
validator s/stage.akk:109	expect InlineDatum(raw_new_datum) = continuing_output.datum	spend_rejects_non_inline_datum
validator s/stage.akk:110	expect new_datum: StageDatum = raw_new_datum	spend_rejects_new_datum_of_the_wrong_type
validator s/stage.akk:125	expect continuing_output.value == own_input.output.value	spend_rejects_ada_drain_while_keeping_the_token
validator s/stage.akk:144	expect list.has(tx.extra_signatories, admin)	spend_rejects_missing_admin_signature, mint_rejects_missing_admin_signature
validator s/stage.akk:150	expect [Pair(asset_name, 1)] = dict.to_pairs(assets.tokens(tx.mint, policy_id))	mint_rejects_two_units_of_the_thread, mint_rejects_two_threads_in_one_tx, mint_rejects_two_asset_names_when_both_are_present_in_the_output, mint_rejects_a_burn

File:line	expect	Tests that abort there
validator s/stage.ak:153	expect [thread_output] = carrying_thread(tx.outputs, policy_id, asset_name)	mint_rejects_token_not_locked_in_the_script, mint_rejects_output_holding_extra_units_of_the_token, mint_rejects_two_outputs_each_carrying_one_unit
validator s/stage.ak:154	expect InlineDatum(raw_datum) = thread_output.datum	mint_rejects_non_inline_datum
validator s/stage.ak:155	expect initial_datum: StageDatum = raw_datum	mint_rejects_initial_datum_of_the_wrong_type

Mutation testing

51 mutants: 51 killed, 0 survived, 0 invalid.

Each row removes or weakens one check the validator uses to reject a transaction — an `and` condition, a boolean `expect`, a list pattern, a `when` branch, or the trailing `fail` — and reruns the suite. "Killed" means at least one test turned red without that check; a mutant that survives means nothing defends it.

File:line	What was removed	Original check	Result	Tests that caught it
validator s/stage.ak:37	filter condition	<code>list.filter(outputs, fn (output) { output.address == address })</code>	killed	<code>spend_accepts_an_unrelated_output_elsewhere</code>
validator s/stage.ak:51	<code>and</code> conjunct	<code>output.address.payment_credential == Script(policy),</code>	killed	<code>mint_rejects_token_not_locked_in_the_script</code>
validator s/stage.ak:52	<code>and</code> conjunct	<code>assets.quantity_of(output.value, policy, stage_ref) == 1,</code>	killed	<code>mint_rejects_output_holding_extra_units_of_the_token</code>
validator s/stage.ak:60	filter condition	<code> > list.filter(fn(input) { input.output.address == address })</code>	killed	<code>spend_accepts_an_extra_wallet_input</code>
validator s/stage.ak:69	accepted an open lower bound	<code>expect Finite(lower) = range.lower_bound.bound_type</code>	killed	<code>spend_rejects_missing_lower_bound</code>
validator s/stage.ak:70	accepted an open upper bound	<code>expect Finite(upper) = range.upper_bound.bound_type</code>	killed	<code>spend_rejects_missing_upper_bound</code>
validator s/stage.ak:72	<code>and</code> conjunct	<code>now >= lower,</code>	killed	<code>spend_rejects_timestamp_one_below_lower_bound</code>
validator s/stage.ak:73	<code>and</code> conjunct	<code>now <= upper,</code>	killed	<code>spend_rejects_timestamp_outside_validity_range, spend_rejects_timestamp_one_above_upper_bound</code>

File:line	What was removed	Original check	Result	Tests that caught it
validator s/stage.ak:91	boolean expect	expect assets.quantity_of(own_input.output.value, own_policy, old_datum.stage_ref,) == 1	killed	spend_rejects_own_input_with_two_units_when_a_decoy_absorbs_carrying_thread
validator s/stage.ak:100	boolean expect	expect inputs_at_addresses(tx.inputs, own_addresses) == 1	killed	spend_rejects_two_script_inputs
validator s/stage.ak:101	list pattern: extra elements	expect [continuing_output] = at_address(tx.outputs, own_address)	killed	spend_rejects_a_second_output_at_the_exact_script_address
validator s/stage.ak:104	list pattern: extra elements	expect [_] = carrying_thread(tx.outputs, own_policy, old_datum.stage_ref)	killed	spend_rejects_thread_split_across_a_staking_variant_of_the_script_address
validator s/stage.ak:107	boolean expect	expect list.has(tx.extra_signatories, admin)	killed	spend_rejects_missing_admin_signature
validator s/stage.ak:125	boolean expect	expect continuing_output.value == own_input.output.value	killed	spend_rejects_ada_drain_while_keeping_the_token
validator s/stage.ak:136	and conjunct	valid_datum_evolution(old_datum, new_datum, to_completion),	killed	spend_rejects_new_datum_of_the_wrong_type, spend_rejects_invalid_transition, spend_rejects_leaving_completed and 4 more
validator s/stage.ak:137	and conjunct	time_ok,	killed	spend_rejects_timestamp_outside_validity_range, spend_rejects_open_ended_validity_range, spend_rejects_timestamp_one_below_lower_bound and 3 more
validator s/stage.ak:144	boolean expect	expect list.has(tx.extra_signatories, admin)	killed	mint_rejects_missing_admin_signature
validator s/stage.ak:150	list pattern: extra elements	expect [Pair(asset_name, 1)] = dict.to_pairs(assets.tokens(tx.mint, policy_id))	killed	mint_rejects_two_asset_names_when_both_are_present_in_the_output
validator s/stage.ak:150	list pattern: any quantity	expect [Pair(asset_name, 1)] = dict.to_pairs(assets.tokens(tx.mint, policy_id))	killed	mint_rejects_a_burn
validator s/stage.ak:153	list pattern: extra elements	expect [thread_output] = carrying_thread(tx.outputs, policy_id, asset_name)	killed	mint_rejects_two_outputs_each_carrying_one_unit
validator s/stage.ak:158	and conjunct	initial_datum.stage_ref == asset_name,	killed	mint_rejects_asset_name_not_matching_stage_ref

File:line	What was removed	Original check	Result	Tests that caught it
validator s/stage. a k:159	and conjunct	valid_initial_datum(initial_datum),	killed	mint_rejects_starting_outside_pending, mint_rejects_preloaded_evidence
validator s/stage. a k:164	fail	fail	killed	else_rejects_other_script_purposes
lib/propn exus/fsm. a k:95	when branch	Pending -> to == InProgress	killed	t_pending_to_pending, t_pending_to_observed, t_pending_to_completed
lib/propn exus/fsm. a k:96	when branch	InProgress -> to == Observed to == Completed	killed	t_in_progress_to_pending, t_in_progress_to_in_progress
lib/propn exus/fsm. a k:97	when branch	Observed -> to == InProgress	killed	t_observed_to_completed, t_observed_to_pending, t_observed_to_observed
lib/propn exus/fsm. a k:98	when branch	Completed -> False	killed	t_completed_is_terminal_to_pending, t_completed_is_terminal_to_in_progress, t_completed_is_terminal_to_observed and 3 more
lib/propn exus/fsm. a k:106	and conjunct	new.project_ref == old.project_ref,	killed	t_identity_rejects_project_swap
lib/propn exus/fsm. a k:107	and conjunct	new.stage_ref == old.stage_ref,	killed	t_identity_rejects_stage_swap, spend_rejects_identity_rewrite
lib/propn exus/fsm. a k:108	and conjunct	new.sequence_order == old.sequence_order,	killed	t_identity_rejects_sequence_swap
lib/propn exus/fsm. a k:109	and conjunct	new.validation_critical == old.validation_critical,	killed	t_identity_rejects_criticality_swap
lib/propn exus/fsm. a k:122	if body	bytearray.length(evidence_root) == commitment_length	killed	t_critical_needs_a_full_commitment, t_evolution_rejects_critical_without_evidence, spend_rejects_completing_critical_without_evidence
lib/propn exus/fsm. a k:136	and conjunct	new.state == to,	killed	t_evolution_rejects_datum_state_mismatch
lib/propn exus/fsm. a k:137	and conjunct	valid_transition(old.state, to),	killed	t_evolution_rejects_leaving_terminal, spend_rejects_leaving_completed
lib/propn exus/fsm. a k:138	and conjunct	identity_preserved(old, new),	killed	spend_rejects_identity_rewrite

File:line	What was removed	Original check	Result	Tests that caught it
lib/propnexus/fsm.akk:143	and conjunct	to == Completed,	killed	t_evolution_rejects_completion_payload_matching_a_non_completing_transition
lib/propnexus/fsm.akk:144	and conjunct	new.evidence_root == evidence_root,	killed	t_evolution_rejects_evidence_root_mismatch_between_redeemer_and_data
lib/propnexus/fsm.akk:145	and conjunct	new.completed_at == now,	killed	t_evolution_rejects_completed_at_mismatch_between_redeemer_and_now
lib/propnexus/fsm.akk:146	and conjunct	now > 0,	killed	t_evolution_rejects_zero_timestamp
lib/propnexus/fsm.akk:147	and conjunct	completion_evidence_ok(old.validation_critical, evidence_root),	killed	t_evolution_rejects_critical_without_evidence, spend_rejects_completing_critical_without_evidence
lib/propnexus/fsm.akk:151	and conjunct	to != Completed,	killed	t_evolution_rejects_completion_without_payload
lib/propnexus/fsm.akk:152	and conjunct	new.evidence_root == old.evidence_root,	killed	t_evolution_rejects_evidence_rewrite_on_flag, prop_non_completing_evolution_preserves_evidence
lib/propnexus/fsm.akk:153	and conjunct	new.completed_at == old.completed_at,	killed	t_evolution_rejects_completed_at_rewrite_on_flag
lib/propnexus/fsm.akk:162	and conjunct	len > 0,	killed	t_initial_rejects_empty_refs
lib/propnexus/fsm.akk:163	and conjunct	len <= max_ref_length,	killed	t_initial_rejects_ref_longer_than_an_asset_name
lib/propnexus/fsm.akk:173	and conjunct	d.state == Pending,	killed	t_initial_rejects_non_pending_state, mint_rejects_starting_outside_pending
lib/propnexus/fsm.akk:174	and conjunct	d.evidence_root == "",	killed	t_initial_rejects_preloaded_evidence
lib/propnexus/fsm.akk:175	and conjunct	d.completed_at == 0,	killed	t_initial_rejects_preloaded_timestamp
lib/propnexus/fsm.akk:176	and conjunct	d.sequence_order > 0,	killed	t_initial_rejects_non_positive_sequence
lib/propnexus/fsm.akk:177	and conjunct	valid_ref(d.project_ref),	killed	t_initial_rejects_empty_refs

